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ORIGINAL PAPER

Baited remote underwater video estimates of benthic fish and invertebrate diversity within the eastern Canadian Arctic

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Abstract

Enhanced survey efforts in the Arctic are currently needed to broaden our understanding of marine species distributions and establish the baseline data and methodologies necessary to detect potential responses to changing regional climate and human activities. For the first time, a baited remote underwater video (BRUV) survey was conducted in the Eastern Canadian Arctic, providing new biological information within data-poor regions of interest for marine conservation and potential fisheries development. A total of 31 camera deployments were conducted in 2015–2016 aboard an industry fishing vessel during summer exploratory fishing cruises, with observations occurring in the marine waters of the Qikiqtaaluk (Baffin) Region of Nunavut. In total, 18 invertebrate taxa and 14 fish taxa were observed at the baited camera, with significant differences in assemblages among sites associated with spatial variation in temperature and depth. The Greenland shark *Somniosus microcephalus* and polar cod *Boreogadus saida* were the dominant fish species observed, with the presence of brittle stars Ophiuroidea, amphipods and chaetognaths dominating invertebrate communities. Comparisons with concurrent fisheries catch data (whelk pots) demonstrated the baited camera's ability to comparably detect local fish taxa when deployed over uniform, fine-grain sediment substrates, while local invertebrate assemblages were often better captured by fisheries surveys. These results illustrate the potential of low-impact BRUV survey methods to efficiently provide baseline data on spatial patterns of diversity and their drivers to advance our understanding of marine ecosystems within understudied polar regions.

Keywords Canadian Arctic · Baited camera · BRUV · Species diversity · Arctic marine communities

Introduction

Effective monitoring and management of Arctic marine ecosystems urgently requires robust biological information of species distributions and abundance estimates. Ongoing climate change in the Arctic at an unprecedented rate has resulted in warming three times faster than the global average (Hoegh-Guldberg and Bruno 2010; Duarte et al. 2012). These observations and projected future changes in temperature and sea ice extent will undoubtedly impact fauna throughout the region, with high potential to change the structure and functioning of Arctic marine ecosystems (Wassmann et al. 2011; Frainer et al. 2017). The Canadian Arctic has also experienced increases in human activities as changing sea ice increases accessibility for shipping, tourism, petroleum and mineral exploitation (Prowse et al. 2009). Fishing effort is expected to grow as fishers follow boreal species expanding to warming northern waters (Cheung et al. 2010; Christiansen et al. 2014). Fishing interests within many northern Arctic communities is also on the rise (Wheeland et al. 2014; Wheeland and Devine 2015; Jacobsen et al. 2018), with communities seeking to explore local waters for resources to expand subsistence fisheries development and potentially join commercial markets. However, as in other emerging fisheries (Anderson et al. 2008), these new fishing grounds are often extremely data poor, with little to no previous sampling in many areas (Coad and Reist 2017). Often these fishing grounds occur in depths beyond the scope of coastal Inuit traditional knowledge.

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Therefore, broadening our understanding of the biogeography of both commercial and non-commercial species requires enhanced survey efforts in the Arctic in order to establish the baseline data and methodologies necessary to detect ecosystem responses to changing regional climate and human activities.

The large geographic area, seasonal or perennial ice cover and high operating costs in remote northern regions have constrained fisheries and ecosystem surveys in Arctic marine waters and limited understanding of the spatial distributions, abundances and functional roles of many species residing in Arctic waters. In the eastern Canadian Arctic, fisheries surveys have largely focused offshore in Baffin Bay and Davis Strait where targeted commercial species such as Greenland halibut *Reinhardtius hippoglossoides* and Northern shrimp *Pandalus borealis* are predominately harvested (Jørgensen et al. 2011). In contrast, few surveys have been conducted within the Canadian Arctic Archipelago (Mueter et al. 2013), with most occurrence data for fishes within this region derived from early Nordic surveys dating back to the late 1800s (e.g. Videnskabs-Selskabet i Kristiania 1907), which biased survey coverage toward southern and coastal regions. Based on these early reports and limited sampling throughout the last century, 207 of the 221 currently known Arctic marine fish species in the Canadian Arctic were reported from Nunavut waters (Coad and Reist 2017). However, the majority of these species lack sufficient data for stock or conservation assessments, with no assessments having occurred for roughly 95% of Arctic marine fish species (Christiansen and Reist 2013). Whether additional undocumented taxa occur in these waters and details of their biogeography throughout the region is unknown. The same inherent sampling limitations in the region leave similar knowledge gaps for invertebrate taxa in the Canadian Arctic, with limited understanding of local assemblages, distributions and relative abundance of invertebrate species (CAFF 2017).

An area of particular concern in the light of changing Arctic conditions are the waters of Tallurutiup Imanga—the Lancaster Sound region—situated at the eastern entrance of the Northwest Passage, with the broader Lancaster Sound ecoregion extending to encompass the Gulf of Boothia and Jones Sound (Spalding et al. 2007). This significant ecoregion is known to support substantial biomass of Arctic wildlife year round. A variety of species significant to Inuit culture for food, social and ceremonial purposes utilise the waterway for vital feeding and nursery grounds, including bowhead, narwhal and beluga whales, seabirds, polar bears, seals and walruses (Darnis et al. 2012; Laidre et al. 2015; Matley et al. 2015). Many species rely on the productivity of this region in the open water season, when rich plankton blooms coincide with ice-retreat and large schools of polar cod *Boreogadus saida* form in near-shore waters (Welch et al. 1993; Hannah et al. 2009). Efforts ongoing since the early 1980s to establish the Tallurutiup Imanga National Marine Conservation Area (NMCA) seek to preserve these resources and protect ecosystems within this region. In September 2017, the Government of Canada defined the boundaries of this prospective NMCA and, pending negotiation, this region could soon become Canada's largest protected area at 109,000 km^2. However, the limited survey data for benthic ecosystems within this ecoregion leaves a major gap regarding the potential importance of this area to species groups beyond marine mammals and seabirds.

The many established survey methods for fish and invertebrate assemblages each bring inherent strengths and limitations. Historically, extractive techniques have dominated monitoring of fishes and other benthic megafauna populations, either using records from commercial catch for target and bycatch species assessment, or through fishery-independent scientific surveys. However, several issues associated with extractive methods limit their desirability for use in certain circumstances, often resulting in trade-offs between detectability and habitat impacts. Gear selectivity can vary across species and sizes (MacLennan 1992; Fraser et al. 2008), leading to inconsistencies and misrepresentation of local assemblages through false absences in surveys. Fixed fishing gears (e.g. longlines and pots) can provide useful data with reduced impact on benthic habitats; however, gear selectivity can strongly bias toward certain species and/or sizes. In contrast, bottom trawl gears can capture a wide variety of species, but potentially destroy benthic habitats in the process (Jones 1992; Thrush and Dayton 2002; Tillin et al. 2006), which is unsuitable for regions harbouring sensitive sessile fauna such as corals and sponges. Likewise, some species experience high capture stress and post-capture mortality rates in both mobile and fixed gear types (Gallagher et al. 2014; Barkley et al. 2017), rendering them inappropriate for certain species, including at-risk species or populations.

As an emerging complimentary or alternative method to traditional survey techniques, baited remote underwater video (BRUV) surveys for ecological monitoring has steadily increased since the 1990s, largely driven by technological progress improving image quality, storage and battery capacity (Mallet and Pelletier 2014). Many shallow-water BRUV platforms can now be assembled with inexpensive and readily available components (De Vos et al. 2015; Watson and Huntington 2016; Bergshoeff et al. 2017). The use of these optical technologies as versatile, non-destructive tools to survey marine environments has gained momentum worldwide, with deployments to date reported from all continents and all the world's oceans (Mallet and Pelletier 2014; Whitmarsh et al. 2016). Increased use has led to variation in methodology and survey design, including differences in bait type and preparation, camera orientation (horizontal or vertical), soak time,

use of a single camera vs. stereo-camera designs (Whitmarsh et al. 2016) and modifications to monitor pelagic ecosystems (Bailey et al. 2007; Letessier et al. 2013; Bouchet and Meeuwig 2015; Jamieson 2016). BRUVs can generate many types of data including benthic habitat characterisation, species and/or functional diversity, body sizes, swimming speed, and facilitate analyses of habitat associations, animal behaviour such as foraging methods and the relative abundances and distributions of identified species (Mallet and Pelletier 2014; Whitmarsh et al. 2016).

These characteristics have made BRUVs useful in surveying sensitive habitats such as marine protected areas (Cappo et al. 2003; Terres et al. 2015), coral reef habitats (Linley et al. 2015; Lindfield et al. 2016) and other habitats conducive to the low-impact nature of BRUV surveys (Henriques et al. 2002; King et al. 2008; Yeh and Drazen 2009). Evidence suggests that the greatest deterioration in seafloor habitat occurs when bottom fishing first begins (Kaiser et al. 2002). However, given comparatively few commercial fisheries in the Arctic relative to most coastal regions (Christiansen et al. 2014; Stock et al. 2017), many benthic marine ecosystems have been spared the impacts of heavy bottom trawling, and therefore continue to support relatively pristine habitats with significant concentrations of cold-water corals and sponges (Kenchington et al. 2011). Consequently, emerging survey efforts should strive to maintain the integrity of these fragile benthic habitats. Significantly fewer BRUV surveys have occurred in polar environments compared to temperate and tropical regions, with only a few studies from Antarctic (Yau et al. 2002; Smale et al. 2007) and Arctic (Premke et al. 2006) waters. Despite the limited use of BRUVs in polar environments to date, their non-destructive quality, cost effectiveness and potential capability of sampling under seasonal sea ice, offer a valuable method for surveying polar ecosystems.

The goal of our study was to provide new data on fish and invertebrate occurrences and explore patterns of diversity in Arctic marine communities using BRUV survey methods in parallel with fisheries sampling methods within the Eastern Canadian Lancaster Sound ecoregion. We identify benthic epifauna to describe local fish and invertebrate assemblages, and compare video-derived diversity estimates to catch data from co-occurring exploratory fishing to assess the detection capacity of optical technologies for future ecosystems surveys in the North. Specifically, we use presence–absence information and relative abundance estimates from video data to analyse differences among regions, including assemblage comparisons along depth and temperature gradients. Given expectations of increased fishing efforts in our study areas with fisheries interests expanding in the north, we demonstrate non-extractive baited cameras as a cost-effective and efficient survey method to improve baseline data for fish and invertebrate communities in remote and deepwater Arctic areas.

Methods

Study area

Camera deployments and exploratory fishing occurred in the following five broad regions within the northern Canadian territory of Nunavut: (1) Admiralty Inlet and within Adams Sound near the community of Ikpiarjuk (hereafter 'Arctic Bay'); (2) Central Lancaster Sound; (3) Southeast McDougall Sound and Barrow Strait near the community of Qausuittuq (hereafter 'Resolute'); (4) Eastern Jones Sound including Starnes Fiord and Grise Fiord near the community of Aujuittuq (hereafter 'Jones Sound') and (5) interior and adjacent waters of Scott Inlet (Fig. 1; Online Resource 1). All data were obtained in collaboration with the Arctic Fishery Alliance (AFA; www.arcticfisheryalliance.com), a 100% Inuit-owned fishing enterprise, and collected aboard their 99-ft fishing vessel *Kiviuq I*. This mutually beneficial science–industry partnership provided real-time scientific catch data analysis to help inform exploratory fishing efforts, while the vessel simultaneously offered a platform of opportunity for researchers to access remote regions to collect valuable data from unstudied and/or poorly sampled waters using complementary research methods.

Exploratory fishing

A total of 104 fixed pot gear fishing sets were completed ($n = 35$ in Arctic Bay, $n = 31$ in Resolute, $n = 38$ in Jones Sound) using a combination of whelk pots [conical, 94 cm base with 13 cm opening, 1–2.2 cm mesh, mean 31.2 pots (± 12.0 SD) per set, mean soak time 27.2 h (± 12.4 SD)] and two sizes of shrimp traps [Small: 82 cm L × 32 cm W × 37 cm H, 0.65 cm mesh; Large: 122 cm L × 62 cm W × 36 cm H, 2.4 cm mesh, mean 23.6 traps (± 2.2 SD) per set, mean soak time 26.8 h (± 14.9 SD)]. Limited deployments of longline gear were also used, with a total of 9 sets ($n = 5$ in Arctic Bay, $n = 1$ in Resolute, $n = 3$ in Jones Sound) with an average soak time of 26.0 h (± 14.9 SD). In 2015, longline strings were comprised of 750 hooks, with equal numbers of sizes 12, 14 and 16 circle hooks. In 2016, longline strings were comprised of 375 hooks and only size 14 circle hooks used for all sets. Hooks on gangions were spaced approximately 1.8 m apart along the length of the groundline (1.5 cm diameter rope). All fishing gear sets were baited with commercial squid bait, and fitted with data loggers (DST centi-TD, Star-Oddi, Gardabaer, Iceland; accuracy ± 0.1 °C) to determine depth and bottom temperature for each set. All catch was identified to the lowest possible

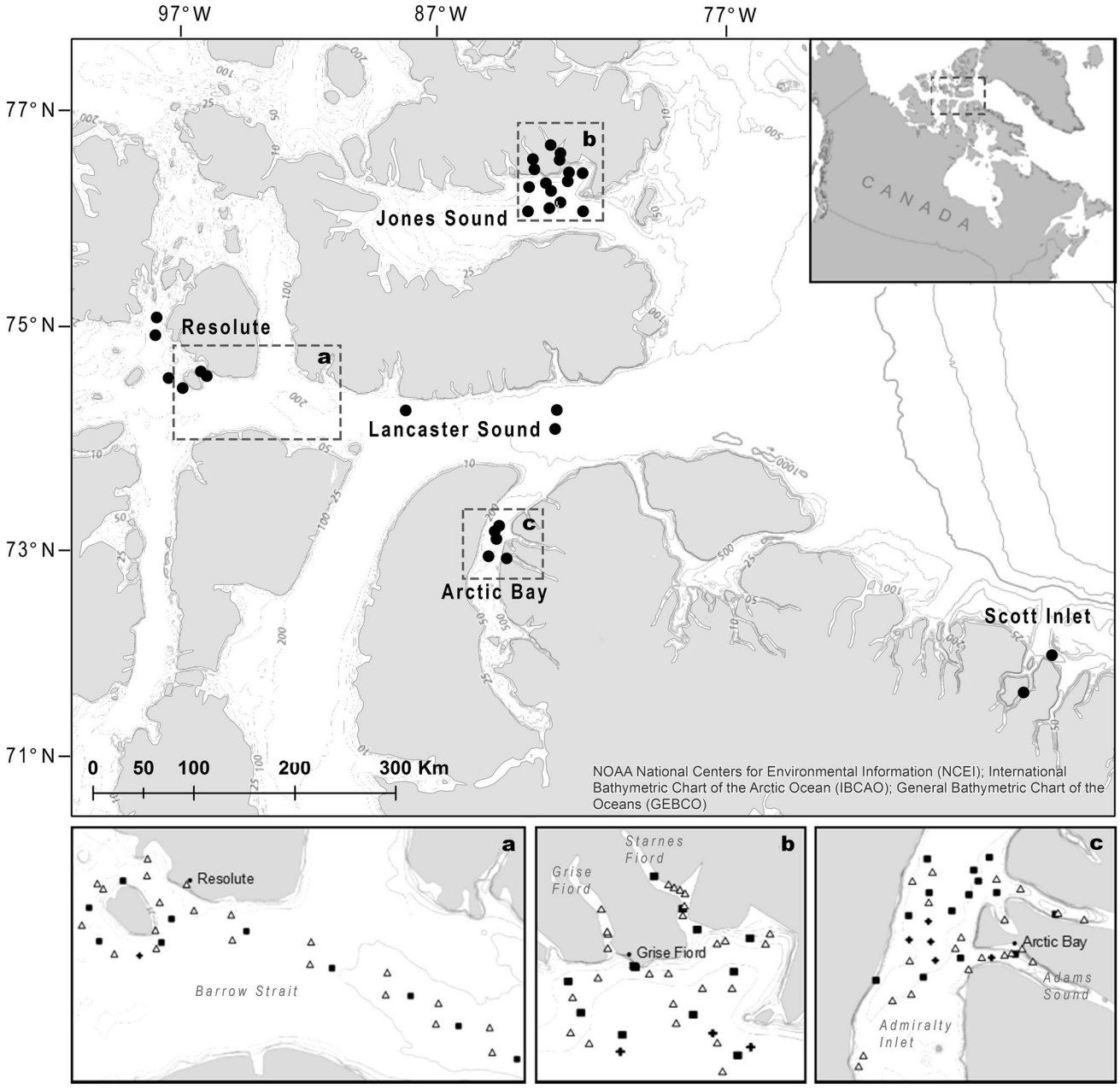

Fig. 1 Map of study area showing positions of each baited camera deployment in the five sampled regions of Arctic Bay, Lancaster Sound, Resolute, Jones Sound and Scott Inlet. Insets show locations of exploratory fishing sets in Resolute (*a*), Jones Sound (*b*) and Arctic Bay (*c*) regions. Gear types include whelk pots (triangles), shrimp traps (squares) and longlines (crosses)

taxonomic level, then counted and weighed to derive abundance and biomass per set.

Baited camera system

A total of 31 baited camera deployments were conducted alongside AFA exploratory fishing during August–September 2015–2016. The baited camera lander consisted of a single high-definition camera with integrated reference lasers (6.24 cm between parallel lasers) and white light source (1Cam Alpha, Aquorea LED; SubC Imaging Inc., Clarenville, Newfoundland and Labrador; www.subcimaging.com) mounted onto a weighted aluminium frame tethered to a surface buoy for retrieval. The camera was positioned at the top of the frame 1.6 m above the seafloor and oriented downward and outward at approximately a 60° angle, with continuous recording at each location (Online Resource 2). A bait arm positioned 50 cm above the seafloor and extending

toward the field of view was baited with 6–8 squid (approx. 2 kg, with half inserted in a bait bag) for each deployment.

The camera system was deployed similar to a series of fishing pots, with surface buoys leading to anchors on either side of the frame, and rope (1.43 cm diameter) of equivalent length to the deployment depth extending from both sides of the frame and leading to each anchor. This design ensured that the commercial hauler aboard the vessel could retrieve the camera system (Online Resource 2). The camera was deployed each morning and retrieved opportunistically the same day dependent on daily fishing activities, with a target deployment duration of 8–10 h. Bottom temperatures at each camera set were derived from depth–temperature data loggers attached either to the nearest bottom fishing gear set (Resolute and Arctic Bay) or from additional DST centi-TD Star-Oddi loggers attached directly on the camera frame (Jones Sound and Scott Inlet). For camera deployments in Lancaster Sound ($n = 3$) where data loggers were unavailable, temperature at equivalent depths was taken from CTD (conductivity, temperature, depth; Seabird 911, Seabird Scientific; accuracy ± 0.001 °C) profiler casts aboard the CCGS Amundsen in August 2015 (ArcticNet 2015) at locations nearest (< 50 nm) to camera deployments.

Video analysis

All fishes and invertebrates were identified to the lowest possible taxonomic level based on morphological characteristics and through consultation with regional taxonomic experts when necessary. Videos associated with each camera set were observed in full, recording first arrival times for all fish taxa, and enumerating a measure of relative abundance (n_{max}), calculated as the maximum number of individuals per species/taxa present within a single video image. This approach ensured counting duplicate individuals did not occur. For *S. microcephalus*, where unique markings (i.e. scar patterns and colouration) made it possible to distinguish and quantify individuals, an additional cumulative measure of local abundance was derived for each deployment (Devine et al. 2018). Fish lengths were estimated using the software ImageJ (Schneider et al. 2012) for all fishes that swam along the seafloor in plane with the camera reference lasers, providing scale for measurements.

For the characterisation of invertebrate communities, a one-minute video clip (starting at minute 0) was analysed at half-hour video intervals throughout each camera deployment, identifying all invertebrates to the lowest taxonomic group possible. The number of video intervals within a set depended on set duration (Table 1), and ranged from 6 to 21. Deployments where water turbidity or an obstruction in the field of view (i.e. rope) impeded a clear view of the seafloor were not included in analyses. Sessile organisms typically occurred throughout a video set—unless an outside force disturbed the substrate and/or camera frame, such as the movements of *S. microcephalus*. A one-minute video clip would therefore capture their presence, and subsequent clips allowed detection of more mobile organisms moving in and out of the field of view (e.g. ophiuroids and gastropods). Observations of taxa within each one-minute interval were accumulated throughout the duration of each set in order to characterise the cumulative invertebrate assemblage for each deployment. Benthic habitat for each set was characterised using descriptions of surface substrate type based on the Wentworth scale (Wentworth 1922), determined using reference lasers for spatial scale. Given the different grades of fine sediment are not readily distinguishable from video footage, the term 'fine-grained sediment' was used to encompass substrate types of clay, silt and/or sand.

Camera-catch detectability comparison

The identities of fish and invertebrate taxa in the videos were compared to those captured in fishing sets conducted in the same area to assess species detection capacity of fishing gear and baited cameras in the waters adjacent to each community. Catch data for these comparisons were collected during exploratory fishing efforts by the AFA near the communities of Arctic Bay, Resolute and Grise Fiord (Jones Sound) (Fig. 1). Catch composition from the nearest (< 30 km) whelk pot string to each individual camera set was compared to species presence in videos to assess species detectability between baited camera and fixed gear survey methods. This comparison excluded large fish species (i.e. *S. microcephalus,* Arctic skate *Amblyraja hyperborea,* and *R. hippoglossoides*) and pelagic invertebrates that could not be captured by the whelk pot fishing gear, as well as infaunal invertebrates that would not be expected to be captured in the video footage. Both whelk pot mesh sizes (1 cm and 2.2 cm) were used in comparisons because species richness did not differ significantly between mesh sizes ($F_{1,24} = 1.87$, $p = 0.18$). For broad comparison between extractive fishing-based and non-extractive camera survey methods, pooled species detection for fishes and invertebrate taxa in each region were compared between observations from all camera deployments versus observations from all fishing gear types used in exploratory fisheries (i.e. whelk pots, shrimp traps and longlines). All methods were carried out under experimental licenses and ethics approval granted by the Department of Fisheries and Oceans Canada and in accordance with experimental protocol approved by the animal ethics committee of Memorial University of Newfoundland.

Statistical analyses

Similarity in fish and invertebrate assemblages based on region, depth and temperature was explored using the

Table 1 Summary of camera set details including location, depth (m), temperature (°C), set duration (minutes) and a qualitative description of the bottom type based on the video footage

Set	Location	Latitude °N	Longitude °W	Depth (m)	Temperature (°C)	Duration (min)	Bottom type
1	Arctic Bay	73.18	−85.68	610	0.3	570	Fine-grained sediment with cobbles
2	Arctic Bay	73.25	−85.71	645	0.3	595	Fine-grained sediment
3	Arctic Bay	73.29	−85.56	671	0.3	587	Fine-grained sediment
5	Arctic Bay	73.00	−85.42	350	0.0	605	Fine-grained sediment with cobbles
6	Arctic Bay	73.03	−85.96	710	0.3	386	Fine-grained sediment
4	Lancaster Sound	74.11	−83.50	677	1.0	615	Fine-grained sediment
7	Lancaster Sound	74.28	−83.37	720	1.0	495	Fine-grained sediment
8	Lancaster Sound	74.37	−88.37	360	−0.2	462	Fine-grained sediment with boulder
9	Resolute	75.10	−97.00	304	−1.2	250	Fine-grained sediment
10	Resolute	74.62	−95.08	112	−1.0	448	Pebble
11	Resolute	74.50	−95.85	209	−1.1	450	Fine-grained sediment with cobble
12	Resolute	74.94	−96.97	264	−1.2	599	Fine-grained sediment
13	Resolute	74.57	−96.35	210	−1.1	414	Fine-grained sediment
14	Resolute	74.66	−95.29	143	−1.1	395	Pebble with boulders
15	Jones Sound	76.31	−82.78	665	0.2	464	Fine-grained sediment
16	Jones Sound	76.24	−82.62	736	0.2	568	Fine-grained sediment
17	Jones Sound	76.50	−82.14	420	−0.1	487	Fine-grained sediment
18	Jones Sound	76.65	−82.42	262	−0.2	570	Fine-grained sediment
19	Jones Sound	76.56	−82.08	352	−0.1	505	Fine-grained sediment
20	Jones Sound	76.38	−81.86	451	0.1	565	Fine-grained sediment
21	Jones Sound	76.36	−81.35	498	0.2	565	Fine-grained sediment
22	Jones Sound	76.30	−81.96	747	0.2	532	Fine-grained sediment
23	Jones Sound	76.12	−82.33	873	0.2	553	Fine-grained sediment
24	Jones Sound	76.02	−81.55	712	0.2	571	Fine-grained sediment
25	Jones Sound	76.09	−82.78	840	0.2	176	Fine-grained sediment
26	Jones Sound	76.08	−83.58	699	0.2	581	Fine-grained sediment
27	Jones Sound	76.30	−83.43	432	0.2	568	Fine-grained sediment
28	Jones Sound	76.45	−83.15	405	−0.1	406	Fine-grained sediment
29	Jones Sound	76.54	−83.17	233	−0.5	447	Fine-grained sediment
30	Scott Inlet	70.89	−71.60	620	1.1	549	Fine-grained sediment with mixed rocks
31	Scott Inlet	71.12	−70.53	802	1.1	570	Fine-grained sediment

statistical software PRIMER 7 (version 7.0.10, Primer-E, Plymouth, UK). A non-metric multidimensional scaling (nMDS, Kruskal and Wish 1978) ordination plot was generated based on a Bray–Curtis similarity matrix constructed from square-root transformed maximum n_{max} values per set for fish taxa for comparison between regions, and using presence–absence data for invertebrate taxa from camera sets. Analysis of similarity (ANOSIM) tests were performed to test for assemblage differences across sampled regions. Individually ranked species contributions to both among-region separation and within-region similarities were further quantified through similarity percentage (SIMPER) routines (Clarke et al. 2014). Finally, a BEST (BIO-ENV) routine (Clarke and Ainsworth 1993; Clarke et al. 2014) based on Spearman's rank correlation was used to explore the relationship between environmental variables of depth and temperature and Bray–Curtis similarity matrices of fish taxa relative abundance and invertebrate presence–absence observed within our sampled regions.

The number of taxa observed in an area was positively related to the number of video intervals from that site (Online Resource 3; *GLM* Taxa observed = 0 + sample size, poisson distribution; *Invertebrates* $z_{24} = 26.26$, $p < 0.001$; *Fishes* $z_{29} = 13.18$, $p < 0.001$). Therefore, extrapolation of accumulation curves was computed in EstimateS (v.9.1.0) using the Bernoulli product model (Colwell et al. 2012) to generate expected taxa richness for a theoretical sample of 21 video intervals (1 video interval = 30 min) at each site, corresponding to the maximum sample size achieved at a single site within the data set. These standardised

extrapolated species accumulation curves were generated for fishes and invertebrates for each deployment, and using mean calculated taxa richness per video interval across all sets to create cumulative curves to determine optimal soak time necessary to capture diversity. Finally, differences in video-derived fish lengths were explored through an analysis of variance (ANOVA) of means between sampled regions, using post hoc Tukey tests to examine pairwise mean comparisons.

Results

Camera deployments ranged from 176 to 615 min (Table 1), with a total of 258 h of video footage analysed. Deployment depths ranged from 143 to 304 m near Resolute, 360 to 720 m in Lancaster Sound, 350 to 710 m near Arctic Bay, 233 to 873 m in Jones Sound and 620 to 802 m in Scott Inlet (Table 1), corresponding to differences in bathymetry among these regions. Bottom temperatures ranged from −1.2 to +1.1 °C, with warmest sets occurring in the deep waters of Lancaster Sound and in Scott Inlet, and coldest sets in the shallow sub-zero waters surrounding Resolute. Depth explained 61% of variation in bottom temperature across the 31 camera deployments, with water temperature increasing at depth (Fig. 2; $r = 0.771$, $n = 127$, $p < 0.001$). Temperature profiles derived from CTD casts taken throughout the survey area indicate the presence of a cold intermediate layer extending approximately 50–200 m at most regions (Online Resource 4). Substrate varied little among deployments, with 77% of sets comprised of uniform, fine-grained sediments. The remaining sets yielded a combination of fine-grained sediments with scattered cobbles or boulders, with the exception of the two sets within Resolute Pass (Sets 10, 14) characterised by pebbles, cobbles and/or boulders (Table 1).

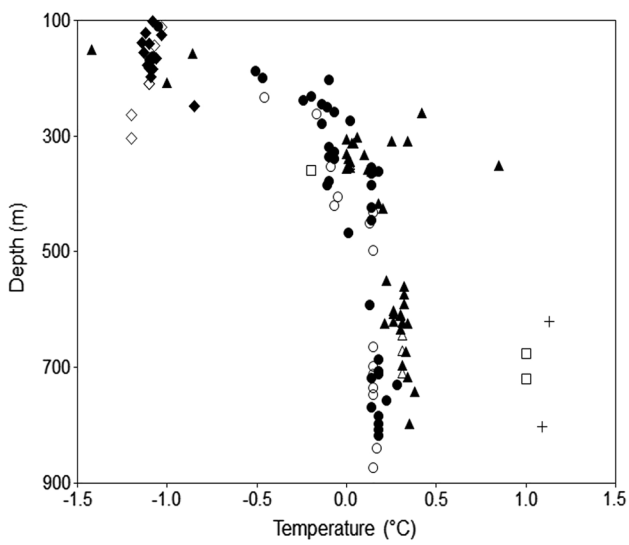

Fig. 2 Bottom depth–temperature relationship across all regions and gear types. All closed symbols correspond to data collected from fishing gear (longlines, pots, traps) sets in Arctic Bay (triangle), Resolute (diamond) and Jones Sound (circle) regions. Open symbols represent camera deployments in these regions. Lancaster Sound (square) and Scott Inlet (cross) regions have camera deployments only

BRUV observation summaries

A total of at least 14 fish species were observed at the baited camera, including 2 elasmobranchs and 12 teleosts, representing 12 different families (Table 2, Fig. 3). *Somniosus microcephalus,* sea tadpole *Careproctus reinhardti* and *B. saida* were the only species observed actively feeding on the squid bait. *Somniosus microcephalus* was the most frequent species observed in our video footage. A total of 142 individuals were distinguished across 25 sets (Online Resource 5), with no individual observed in multiple camera sets. This species occurred in all sampling locations, though in highly variable abundance. At least 18 individuals occurred in a single set in Admiralty Inlet, in contrast to only 3 individuals near Resolute, all in the two shallowest sets within Resolute Pass (see also Devine et al. 2018).

Spatial patterns in distribution and relative abundance were apparent for several fish taxa (Online Resource 6). *Reinhardtius hippoglossoides* was present only in the two deeper deployments in Lancaster Sound (Sets 4 and 7) and one set just outside Scott Inlet (Set 31). For all of these observations, depths were > 650 m and bottom temperatures exceeded 1° C. Similarly, *A. hyperborea* and *C. reinhardti* occurred only in deep sets > 450 m in the open waters of Admiralty Inlet near Arctic Bay, Lancaster Sound and Jones Sound. *Boreogadus saida* was the most abundant teleost and the only species present in all 5 regions, with the highest n_{max} observed in sets within small fiords and inlets (i.e. Grise Fiord and Starnes Fiord in Jones Sound; Adams Sound near Arctic Bay). *Boreogadus saida* occurred in all sets from 200–670 m and was the only species found throughout the full range of temperatures sampled. Sculpins Cottidae spp. and Atlantic poacher *Leptagonus decagonus* were both present only in Arctic Bay and Jones Sound regions, predominately within small fiords at depths 230–700 m and temperatures near 0 °C. Eelpouts *Lycodes* spp. and snailfishes *Liparis* spp. occurred at a wide range of depths and temperatures, from shallow sub-zero waters to warmer deep sets, and were the only other teleosts present near Resolute. Although several species were likely present, morphological traits distinguishing species within the genera *Lycodes* and *Liparis* and family Cottidae are exceedingly subtle, precluding identification to species level through video analysis. Rare or uncommon species (i.e. limited observations within the videos) included American plaice *Hippoglossoides*

Table 2 Summary of fish taxa camera observations per region, reporting mean n_{max} for each region, total number of measurements taken (n_{TL}) and mean total length (cm) ± standard deviation with length ranges in parentheses below

Fish taxa	Family	Arctic Bay $n=5$			Lancaster sound $n=3$			Resolute $n=6$			Jones sound $n=15$			Scott Inlet $n=2$		
		n_{max}	n_{TL}	Mean ± SD (range)	n_{max}	n_{TL}	Mean ± SD (range)	n_{max}	n_{TL}	Mean ± SD (range)	n_{max}	n_{TL}	Mean ± SD (range)	n_{max}	n_{TL}	Mean ± SD (range)
Somniosus microcephalus	Somniosidae	1.8	25	254.5 ± 32.0 (195–314)	1.3	7	235.6 ± 47.8 (157–286)	0.3	3	281.3 ± 28.3 (264–314)	1.3	52	249.5 ± 33.2 (195–320)	1	7	198.3 ± 73.8 (131–325)
Amblyraja hyperborea	Rajidae	0.2	3	20.4 ± 7.6 (15–29)	1.3	49	49.6 ± 11.6 (15–80)	0	–	–	0.9	28	44.0 ± 14.6 (10–76)	0	–	–
Boreogadus saida	Gadidae	1.4	21	14.0 ± 5.0 (9–26)	0.3	25	13.3 ± 2.5 (9–19)	0.8	71	13.4 ± 2.5 (8–21)	1.6	38	15.4 ± 4.8 (7–30)	0.5	–	*
Careproctus reinhardti	Liparidae	0.2	5	20.5 ± 1.5 (19–22)	0.3	1	19	0	–	–	0.4	10	13.3 ± 3.2 (10–16)	0	–	–
Cottidae spp.	Cottidae	0.4	1	12	0	–	–	0	–	–	0.9	19	17.0 ± 3.9 (9–26)	0	–	–
Leptagonus decagonus	Agonidae	0.8	2	22.0 ± 7.1 (17–27)	0	–	–	0	–	–	0.8	7	14.7 ± 2.0 (12–17)	0	–	–
Liparis spp.	Liparidae	1.0	5	18.1 ± 1.3 (17–20)	0	–	–	0.3	2	8.5 ± 2.1 (7–10)	0.1	1	7	0	–	–
Lycodes spp.	Zoarcidae	0.8	4	31.3 ± 14.9 (17–46)	0.7	1	35	0.2	2	12.0 ± 2.8 (10–14)	0.4	5	18.6 ± 5.8 (10–25)	0	–	–
Reinhardtius hippoglossoides	Pleuronectidae	0	–	–	0.7	8	47.4 ± 6.7 (40–62)	0	–	–	0	–	–	0.5	1	51
Mallotus villosus	Osmeridae	0	–	–	0.3	–	*	0	–	–	0	–	–	0	–	–
Eumicrotremus spinosus	Cyclopteridae	0	–	–	0	–	–	0	–		–0.1	1	8	0	–	–
Gymnelus spp.	Zoarcidae	0	–	–	0	–	–	0	–		–0.1	2	16.6 ± 1.9 (15–18)	0	–	–
Hippoglossoides platessoides	Pleuronectidae	0	–	–	0	–	–	0	–		–0	–	–	0.5	2	33.3 ± 7.1 (28–38)
Brosme brosme	Lotidae	0	–	–	0	–	–	0	–		–0	–	–	0.5	1	59
Unknown	Unknown	0	–	–	0	–	–	0	–		–0	–	–	0.5	–	*

The * denotes sets where taxa were present but no individuals measured

As individuals of most species are indistinguishable, length sample sizes do not reflect total number of individuals observed at the camera

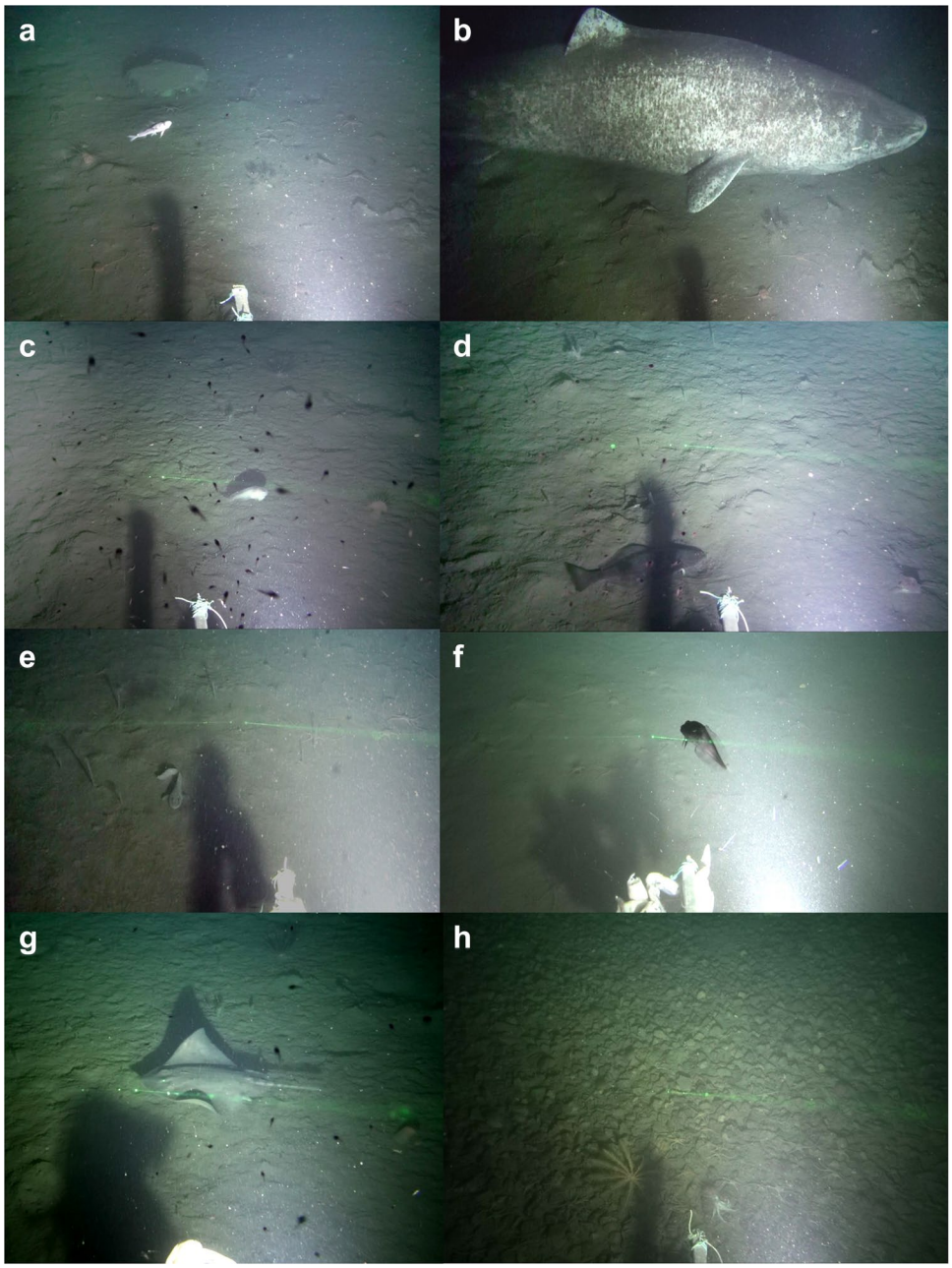

Fig. 3 Images captured by baited camera throughout the study area: **a** *Boreogadus saida*; **b** *Somniosus microcephalus*; **c** *Careproctus reinhardti*; **d** *Reinhardtius hippoglossoides*; **e** *Lycodes* spp.; **f** *Liparis* spp.; **g** *Amblyraja hyperborea*; **h** example of rocky substrate in Resolute with crinoid *Heliometra* spp.

platessoides, fish doctor *Gymnelus* spp., cusk *Brosme brosme,* spiny lumpsucker *Eumicrotremus spinosus,* capelin *Mallotus villosus* and a single pelagic unidentified fish in Scott Inlet (Set 31).

A total of 18 invertebrate taxa were observed within the camera sets (Table 3). Amphipods, chaetognaths and ophiuroids occurred in all camera sets. Nephtheid soft corals were present in all regions and across the full range of sampled depths (112–873 m). Despite high catch frequency of the sea pen *Umbellula encrinus* (8 from longlines, 194 from pots/traps combined), this species was present in just a single camera deployment in Jones Sound (Set 24). Actiniarians were common, present in all locations except Resolute, whereas urchins *Strongylocentrotus* spp. occurred only in a single set in Resolute comprised of pebble substrate. Crinoids *Heliometra* spp. occurred in the shallowest sets from Resolute and Lancaster Sound, both with benthic habitats containing pebble or boulders. Asteroidea sea stars occupied a wide range of depths and were present in all regions except Scott Inlet, whereas the basket star *Gorgonocephalus* spp. were present in only a single set from each of Arctic Bay, Resolute and Jones Sound. The camera was also able to detect the presence of pycnogonid sea spiders in Resolute and Jones Sound, and holothurian sea pigs of the genus *Elpidia* in several sets within Arctic Bay and Lancaster Sound, despite their small size and cryptic nature. Regarding species

Table 3 Summary of invertebrate taxa observed from camera deployments within each region

Region		Arctic Bay	Lancaster sound	Resolute	Jones sound	Scott Inlet	Total
Number of sets		5	3	4	11	2	25
Phylum	Taxon	(1 = present, 0 = absent)					% occurrence (all sites)
Arthropoda	Amphipoda	1	1	1	1	1	100
Chaetognatha	Chaetognatha	1	1	1	1	1	100
Echinodermata	Ophiuroidea	1	1	1	1	1	100
Cnidaria	Nephtheidae	1	1	1	1	1	68
Echinodermata	Asteroidea	1	1	1	1	0	56
Arthropoda	Caridea	1	1	0	1	1	56
Annelida	Polychaeta	1	1	0	1	1	52
Mollusca	Buccinidae	1	1	1	1	1	48
Cnidaria	Actiniaria	1	1	0	1	1	40
Arthropoda	Euphausiacea	1	0	0	1	0	28
Arthropoda	Pycnogonidae	0	0	1	1	0	16
Echinodermata	*Elpidia* sp.	1	1	0	0	0	16
Echinodermata	*Gorgonocephalus* spp.	1	0	1	1	0	12
Bryozoa	Bryozoa	1	0	1	0	0	12
Echinodermata	*Heliometra* spp.	0	1	1	0	0	8
Porifera	Porifera	1	0	1	0	0	8
Cnidaria	*Umbellula encrinus*	0	0	0	1	0	4
Echinodermata	*Strongylocentrotus* spp.	0	0	1	0	0	4

Presence or absence is reported for each region (sets combined by region), and the total percent occurrence within all sets (all regions combined) reported for each taxa

of interest for local emerging fisheries, whelk (family: Buccinidae) were present across all five regions, whereas caridean shrimp were present in all regions except Resolute.

Statistical analyses

Extrapolated invertebrate taxa richness within sets ranged from 4 to 11, with an overall median richness of 7 taxa. Taxonomic richness extrapolated within individual sets was not significantly related to depth or temperature, and did not vary significantly among sampled regions ($z = -11.127$; $p = 0.76$). Extrapolated fish taxa richness ranged from 1 to 6, with an overall median richness of 3. Fish taxonomic richness was also not significantly related to depth or temperature, nor did it vary significantly among sampled regions ($z = -16.061$; $p = 0.81$). Pooled cumulative proportions of observed taxa for all deployments using mean extrapolated richness values indicated variability in theoretical observation time necessary for maximum species detection between invertebrate and fish taxa (Fig. 4). Extrapolated richness curves to 10.5 h (21 video intervals; the maximum number of intervals within a set in this study) deployment time suggest observations

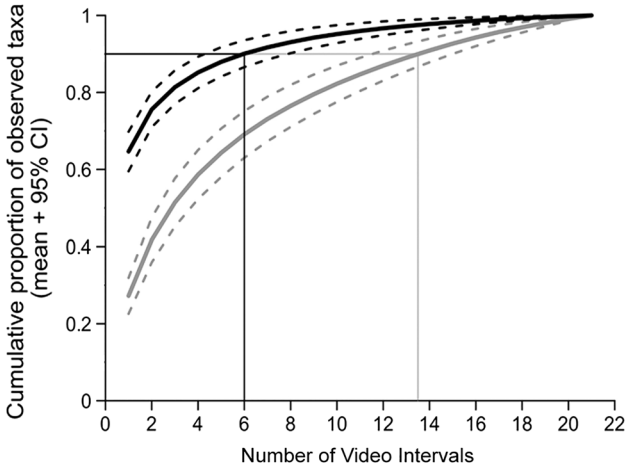

Fig. 4 Combined extrapolated rarefaction curves using mean values for all sets and regions, displayed as cumulative proportion of total observed taxa extrapolated to 21 video intervals (1 video interval = 30 min) for invertebrate taxa (black line) and fish taxa (grey line) diversity with 95% confidence intervals (dashed lines). Vertical lines denote average time by which 90% of taxa were observed

of 90% of invertebrate taxa within the first 3 h (6 video intervals) of deployments; however, 90% of fish taxa were not observed until approximately 6.75 h (13.5 video intervals) (Fig. 4). Although some variability in detection rates occurred between sampled regions, there was strong overlap at the time necessary to detect 90% of taxa (Online Resource 7).

Non-metric multidimensional scaling (nMDS) plots using a Bray–Curtis similarity matrix of square-root-transformed n_{max} for fishes and presence–absence for

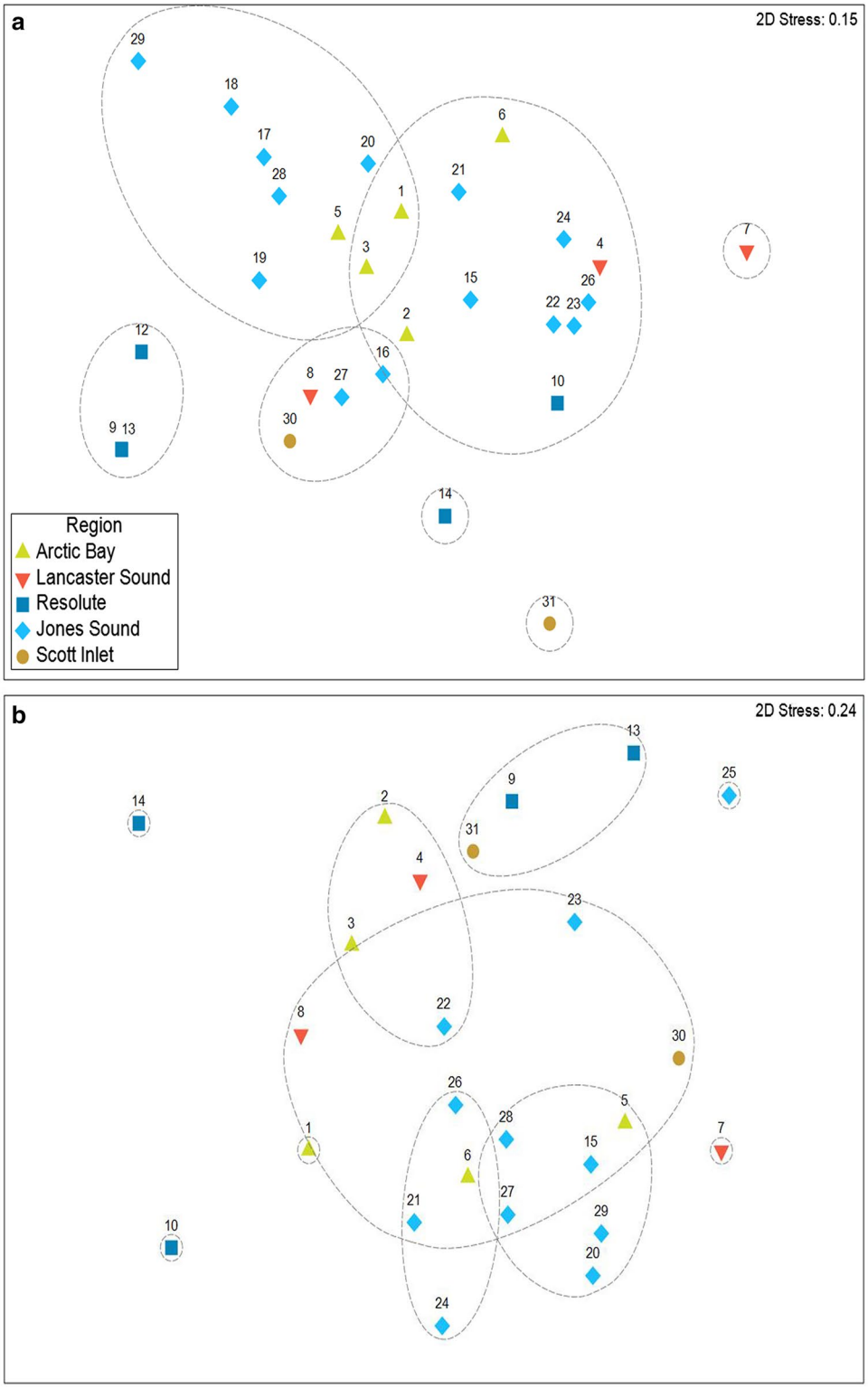

Fig. 5 Non-metric MDS plots of square-root transformed Nmax Bray–Curtis similarity matrices of **a** fish n_{max} and **b** invertebrate P-A from each region. Dotted lines correspond to 50% similarity for fishes and 75% similarity for invertebrates based on hierarchical cluster analyses

invertebrates indicate varying degrees of similarity in marine communities among regions (Fig. 5). ANOSIM analyses of fish relative abundance using n_{max} revealed significantly distinct groups based on region (Global $R = 0.273$, $p = 0.004$). Pairwise comparisons, however, indicate significant differences in assemblages only between Arctic Bay–Scott Inlet ($p = 0.048$), Resolute–Jones Sound ($p = 0.003$) and Jones Sound–Scott Inlet ($p = 0.033$). However, small sample sizes from Scott Inlet ($n = 2$) and Lancaster Sound ($n = 3$) limited power to detect significant differences via permutation tests. SIMPER analysis indicated *S. microcephalus* was the main contributor (33.9–40.2%) to within-group similarities at Arctic Bay, Lancaster Sound, Jones Sound, and it contributed 100% at Scott Inlet. In Resolute, *B. saida* accounted for 83.3% of within-group similarity. BEST (BIO-ENV) routine analysis found the strongest relationship between fish assemblage patterns and the combination of both deployment depth and bottom temperature across sampled regions (Global test $\rho = 0.516$, $p = 0.01$).

Presence–absence ANOSIM analysis of invertebrate communities between regions indicates significant separation of groups (Global $R = 0.27$, $p = 0.006$). Pairwise comparisons further indicate significant differences in assemblages between Lancaster Sound vs. Jones Sound ($p = 0.038$) and Resolute vs. Jones Sound ($p = 0.001$). SIMPER analysis indicates that amphipods, chaetognaths and ophiuroids were the three primary contributors to within-group similarities at Arctic Bay, Lancaster Sound, Jones Sound and Resolute (57.5%, 69.6%, 56.3% and 79.2%, respectively). At Scott Inlet, amphipods, anemones and chaetognaths each contributed 25% to within-group similarity, with the lowest overall dissimilarity observed between Lancaster Sound and Scott Inlet regions (28.7%). Similar to fish taxa, BEST routine analysis also found the strongest relationship between invertebrate presence–absence patterns and the combination of both deployment depth and bottom temperatures (Global test $\rho = 0.297$, $p = 0.01$).

Differences in mean lengths between regions were also observed for several fish taxa (Table 2). Mean lengths of *S. microcephalus* differed significantly among regions ($F_{4,89} = 3.64$, $p = 0.008$), primarily due to smaller sharks observed within Scott Inlet. *Boreogadus saida* sizes were similar between regions, with a slight difference in means observed between Jones Sound and Resolute ($F_{3,151} = 3.162$, $p = 0.02$). Significantly smaller individuals of *A. hyperborea* in Arctic Bay compared to other regions ($F_{2,77} = 8.357$, $p < 0.001$) contrasted significantly larger mean lengths of *L. decagonus* ($F_{1,7} = 7.663$, $p = 0.027$), *Liparis* spp. ($F_{2,5} = 43.49$, $p < 0.001$) and *C. reinhardti* ($F_{2,13} = 13.23$, $p < 0.001$) observed within Arctic Bay camera sets compared to other regions.

Camera-catch data comparison

Comparisons of fish species detection (excluding *S. microcephalus*, *A. hyperborea* and *R. hippoglossoides*) between camera and catch data where the two sampling methods overlapped (near Arctic Bay, Resolute and Jones Sound) yielded 21 incidences where a whelk pot fishing set and BRUV deployment occurred within 30 km of each other. Mean detection proportions among regions found 48% of taxa observations occurred in both camera and catch data, 38% of observations occurred only in camera data and 14% of observations occurred only in catch data (Figs. 6, 7). Across all sampled regions, both methods recorded the presence of *B. saida, C. reinhardti, Cottidae* spp., *E. spinosus, Liparis* spp. and *Lycodes* spp., and the video detected the additional presence of *Gymnelus* spp. and *L. decagonus* that did not occur in corresponding catch data.

In comparing invertebrate composition between camera and catch data where survey methods overlapped ($n = 16$), mean detection proportions among regions found 63% of taxa were sampled using both methods; however, only 7% of observations occurred solely in the camera data, compared to 30% of observations only in catch data (Figs. 6, 7). A majority of invertebrate taxa were detected by both methods across all sampled regions, and both video and catch data confirm the ubiquity of Asteroidea, Ophiuroidea and Buccinidae in our sampled regions. However, the presence of Holothuroidea (non-Elasipodida), Isopoda, *U. encrinus* and the octopus *Bathypolypus arcticus* only occurred within the catch data and were not detected by the camera.

Comparisons of pooled taxa (including large fish and pelagic invertebrates excluded from nearest whelk pot comparisons) for all camera deployments versus all fishing gear sets (i.e. whelk pots, shrimp traps and longlines) within each region show similar trends as whelk pot comparisons for fish and invertebrate detection (Online Resource 8). For pooled

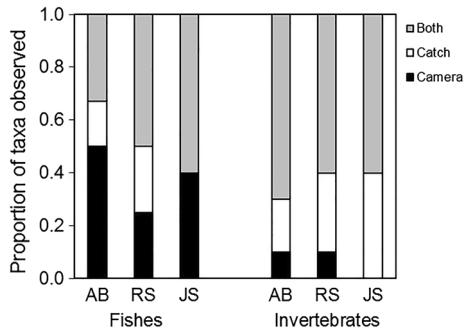

Fig. 6 Proportion of fish and invertebrate taxa observed by both survey methods, only within camera data and only within catch data from nearest whelk pot fishing sets across all regions where spatial overlap of the two survey methods occurred (Arctic Bay, Resolute and Jones Sound)

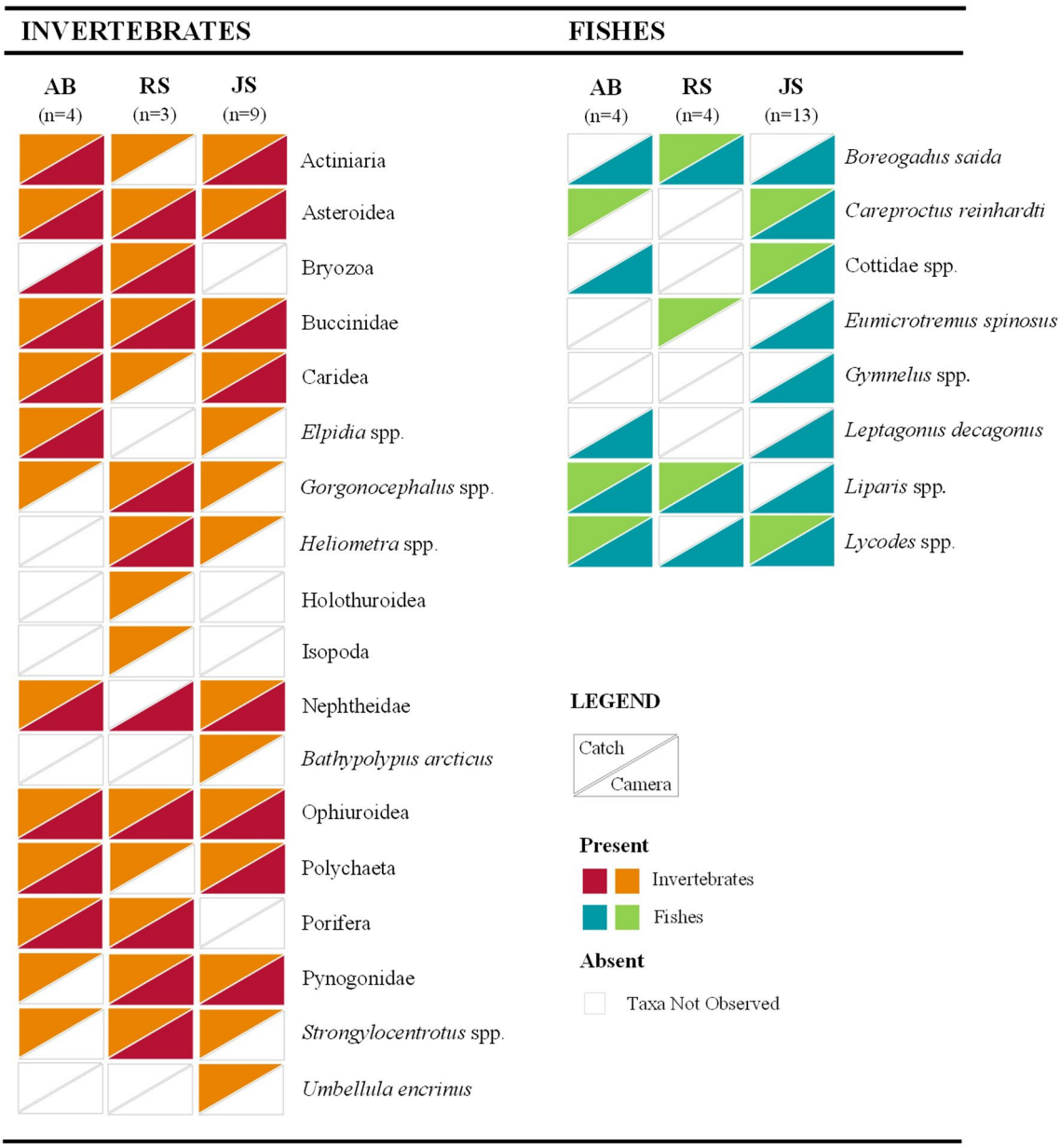

Fig. 7 Fish and invertebrate taxa observed by the camera and nearest whelk pot catch data where spatial overlap of the two survey methods occurred (Arctic Bay (AB), Resolute (RS) and Jones Sound (JS)). White indicates when taxa were not observed. (Color figure online)

fish taxa comparisons, the camera and catch detected the same taxa overall across all sampled regions; however, taxa detection differed within each region. In Arctic Bay, the same 8 taxa of fish were detected by 5 camera sets and by 40 fishing sets (21 whelk pots, 14 shrimp traps, 5 longlines). Similarly, in Jones Sound 15 camera sets detected 10 taxa compared to 9 taxa from 41 fishing sets (26 whelk pots, 12 shrimp traps, 3 longlines), with cameras detecting the additional presence of *E. spinosus* and *Gymnelus* spp., whereas longlines captured *R. hippoglossoides* not observed in camera sets at this location. In Resolute, 6 camera sets detected 4 taxa compared to 7 taxa from 32 fishing sets (21 whelk pots, 10 shrimp traps, 1 longlines). Here the camera also detected the presence of *S. microcephalus* (not present in fishing sets likely due to only a single longline deployment), whereas catch data provided additional observations of *Gymnelus* spp., *Cottidae* spp., *E. spinosus* and *C.reinhardti*.

Pooled invertebrate taxa comparisons for all camera sets and fishing gear types across all regions found 17 taxa detected in both survey methods, with fishing gears detecting the presence of an additional 8–13 taxa in each region (Table S2). In Arctic Bay, only the camera detected the

presence of euphausiids, whereas the catch data detected an additional 8 taxa, including *Strongylocentrotus* spp., *U. encrinus* and *Heliometra* spp. In Resolute, all taxa in the video footage were also present in catch data, with the fishing sets detecting an additional 9 taxa, including anemones, holothurians, caridean shrimp and several infaunal taxa. The highest discrepancy between sampling methods occurred in Jones Sound, with catch data documenting an additional 13 taxa, including *Strongylocentrotus* spp., *Heliometra* spp. crinoids, sponges and the octopus *Bathypolypus arcticus*, while the camera only detected the additional presence of euphausiids. Unsurprisingly, video better detected smaller-bodied, pelagic taxa (Amphipoda, Chaetognatha, Euphausiids) that were not efficiently captured given the mesh sizes of the fishing gear.

Discussion

This study presents the first BRUV survey of fishes and invertebrates in the eastern Canadian Arctic, characterising the benthic environment and taxa of widely ranging sizes within this poorly studied, but rapidly changing, region. The video data described local ichthyofauna, invertebrate communities and habitat composition, with depth and temperature variations explaining significant differences among sampled regions. Comparison of video observations to proximate fisheries catch data demonstrates the capability of baited cameras to be used as an independent or complementary survey method for monitoring biodiversity; however, our BRUV platform better detected local fish taxa compared to invertebrate taxa. Overall, these results demonstrate the potential for BRUV surveys as viable method for monitoring species distributions—particularly fishes—in sensitive areas of the North.

Variability in fish and invertebrate assemblage composition among regions resulted from differences in depth and bottom temperature, with temperatures increasing with depth. Pairwise comparisons of species contributions to between region similarities using fish relative abundance (n_{max}) show lowest dissimilarity between Arctic Bay and Jones Sound (50%). These two regions offer similar temperature and depth ranges, and substrate dominated by fine-grained sediments. Highest dissimilarity occurred between Lancaster Sound and Resolute (22%) and Resolute and Scott Inlet (25%). Temperatures were warmest in Scott Inlet and deep sets of Lancaster Sound, and sets within Resolute generally sampled shallower, colder habitats, potentially contributing to higher species dissimilarity among regions. Invertebrate assemblages were less dissimilar overall among locations, partly due to the presence of amphipods, chaetognaths and ophiuroids in all sets. Lancaster Sound versus Scott Inlet were the least dissimilar, reflecting similar temperature and depth regimes.

Comparisons of pooled fish taxa for all camera deployments versus all fishing gear sets (i.e. whelk pots, shrimp traps and longlines) within a region highlight the potential of BRUV surveys. Variable detectability over different substrate types presumably explains the majority of differences in taxa detection where the two surveys methods overlapped. Where fine-grained sediment bottom was the dominant substrate type, as in Arctic Bay and Jones Sound, the baited camera demonstrated comparable taxa detection capacity compared to fishing catch. Where rocky substrates occur, as in Resolute, small and cryptic species may be concealed from view. Over these substrates, BRUV surveys may be less suitable for characterising local assemblages. Invertebrate taxa detection through video observations was notably lower than fishing catch data, possibly because of smaller invertebrate sizes and behaviour. Overall, the camera detected the presence of most mobile invertebrate macro epifauna collected in fishing gear sets, and also detected euphausiids that were absent in the catch data. Of the invertebrates only observed within the catch data, 40% were infaunal taxa. Camera detection of invertebrates may be improved by additional camera deployments that could help to detect 'rare' fauna and increase encounters with sessile fauna, or an additional downward-oriented camera or small baited trap at the frame base to improve detection of small, cryptic species. As each string of pot gear samples a substantially larger effective area compared with the camera field of view, the pots likely encounter more species, particularly sessile or slow-moving invertebrates.

Both the fishing gears and baited camera used here are fixed gears that rely on the mobility of species for detection. Although catchability issues related to organism size, gape or behaviour may prevent capture within pots or longlines, baited cameras can readily observe a wide range of fauna sizes. Baited cameras do have their own inherent limitations and potential detection biases. Fauna observed at baited cameras may not represent the entire community but rather only 'bait-attending' species (Jamieson 2016); however, baited cameras may attract higher numbers of scavengers and predators without diminishing the abundance of herbivorous or omnivorous species (Harvey et al. 2007). Additionally, some taxa may be attracted not to the bait but to other sources, such as light, noise or presence of prey species, associated with the platform (Jamieson 2016). In contrast, studies have documented taxa-specific avoidance behaviours related to light source, including illumination length (Bailey et al. 2005) or light colour (Raymond and Widder 2007), and bait composition (Wraith et al. 2013; Walsh et al. 2017). Other environmental variables, such as current speed or turbidity, could also influence species behaviour, mobility

and/or odour dispersal and subsequently camera detection capacity.

Rarefaction curves of extrapolated invertebrate taxa richness indicated that the first video interval (i.e. 30 min) characterised an average 65% of the regional taxa diversity, and the first 6 video intervals recorded 90% of the taxa diversity at a site. These results suggest that a set duration of approximately 3 h reasonably characterises benthic invertebrate community diversity. Similar curves of extrapolated fish taxa richness suggest a longer set duration in order to assess regional fish communities, with an average 90% of fish taxa diversity observed at sets within approximately 6–7 h. However, curves for fish diversity extrapolated to 11 h had not yet reached an asymptote, indicating that deployments exceeding 6 h could potentially detect additional taxa. Deployment times of ≤ 60 min characterise a majority of BRUV surveys conducted to date, although many of these shorter deployments targeted depths < 100 m (Whitmarsh et al. 2016). Reduced abundance and potentially diversity at greater depths may require significantly longer soak times and presumably increased replication in order to adequately characterise local assemblages. As such, over 60% of studies conducted to-date sampling below 100 m utilised soak times longer than 90 min (Whitmarsh et al. 2016), with many deep-sea studies often exceeding 11 h (Jones et al. 2003; Sweetman et al. 2014; Jamieson et al. 2017). The Arctic may host lower diversity compared to lower latitudes (e.g. latitudinal diversity gradient, Rex et al. 1993), therefore polar regions may require longer deployments akin to those needed for deep-ocean diversity characterisation. Likewise, the prevalence of non-scavenging species at shallower depths may reduce bait response, potentially requiring more deployments to detect local species, although attraction may be a combination of photo- and chemotaxis for many species.

The use of remote underwater video in marine ecology studies has increased with advancements in optical technologies over the last sixty years. These methods offer well-documented advantages (Mallet and Pelletier 2014; Whitmarsh et al. 2016), including a cost-effective and efficient sampling method geared toward a low-impact, ecosystem-based approach to species conservation and management. This non-extractive method provides valuable fishery independent data suitable for a wide variety of habitats, depths and taxa. However, increased popularity has led to diversification of experimental designs. This lack of consistent protocol for methods, variability in bait type and quantity, as well as uncertainty surrounding the area and longevity of attraction currently limit baited camera methods (Harvey et al. 2013; Whitmarsh et al. 2016) that warrant attention from future studies. Despite these issues, BRUV surveys offer a desirable alternative or complement to traditional survey methods, with wide success in ecological monitoring of a variety of coastal and deep-sea habitats (King et al. 2008; Jamieson et al. 2009; Mallet and Pelletier 2014; Whitmarsh et al. 2014) and protected areas worldwide (Langlois et al. 2006; Bond et al. 2012; Roberson et al. 2015). Significantly fewer BRUV surveys have occurred in polar waters compared to temperate and tropical regions, with a few exploratory studies in the Antarctic (Collins et al. 2002; Gregory et al. 2017) and Arctic waters east of Greenland (Linley et al. 2015). Limited commercial fisheries in the Arctic Ocean (Christiansen et al. 2014; Stock et al. 2017) have spared many benthic marine ecosystems the impacts of bottom trawling, presumably preserving pristine benthic habitats. As northern waters become increasingly more accessible, BRUVs provide an attractive low-impact method to monitor species distributions in a changing polar environment while maintaining the integrity of these sensitive habitats.

Understanding current ecosystem dynamics and predicting potential shifts in local assemblages with future change requires further exploration of biogeographical patterns of species distributions. However, such data are exceedingly sparse for much of the Arctic, and particularly for deep-water environments where warmer water temperatures may support northern range expansions of southern species. For many years, warming in the Arctic has exceeded the global average (Hoegh-Guldberg and Bruno 2010; Duarte et al. 2012). Recent studies already report range shifts for marine fishes, with expansion of boreal species and retraction of Arctic fish communities in the Barents Sea (Fossheim et al. 2015), and nearly two-thirds of North Sea marine fishes have shifted in distribution over the past 25 years, with all but one species expanding northward (Perry et al. 2005). Detecting the presence and rate at which such range shifts occur offer a valuable tool for measuring the impact of climate change on local marine communities; however, detecting such changes requires sufficient time-series data—either from long-term scientific monitoring or historical catch data from commercial fisheries. Given the observed fish taxa detection capacity of BRUVs deployed over uniform, fine-grained sediments—which is the dominant substrate across many deepwater Arctic regions (Andrews et al. 1991)—these platforms may provide a non-intrusive tool for ecological monitoring of a variety of ichthyofauna throughout the Arctic.

High operating costs, inaccessibility due to seasonal ice coverage and limited fisheries exploitation in the north likely explain significant gaps in sampling coverage and temporal monitoring throughout much of the Canadian Arctic (Mallory et al. 2018). Cost-effective and efficient BRUV sampling may provide a suitable conservation-sensitive approach to monitoring Arctic waters and detecting indicators of change. Although previous studies documented most of the fish observed in our study within the Canadian Arctic, the observation of cusk *Brosme brosme* in Scott Inlet likely represents a new record for this area, given that current

published species lists (Coad and Reist 2017) and online distribution resources (OBIS, Fishbase) lack records for this species beyond Davis Strait. Our camera observation potentially expands the record for this fish by over 700 km. With sparse or no previous sampling of fish populations in our study areas, our observations help to extend or fill in major spatial gaps in the range for nearly all species encountered. This contribution demonstrates the utility of optical technologies such as BRUV surveys to monitor marine environments efficiently and detect species shifts in Canada's changing northern waters.

Many habitat-forming, sessile organisms such as corals and sponges are fragile and slow-growing, so removal by fishing gear may significantly impact local populations (Clark et al. 2016). There should therefore be high incentive to use non-destructive tools—such as remote cameras—in pristine, undisturbed regions throughout the high Arctic, to address critical knowledge gaps while maintaining habitat integrity. Our exploratory fishing surveys utilised only fixed gears (pots, traps); however, these gear types captured a surprisingly high numbers of corals: 202 *U. encrinus*, 5 *Virgularia* sp. sea pens, 96 Nephtheid soft corals (including 5 *Gersemia rubiformis* colonies) and 18 sponges from 5 longlines, 14 shrimp traps and 35 whelk pots. Bycatch rates of *U. encrinus* were usually high, as this species frequently occurred in clusters of 5–10 individuals draped on top of each pot/trap, presumably raked off the bottom by rope lines connecting subsequent pots/traps. Life history studies of this species indicate slow growth and high longevity; one of our bycatch specimens measured 230 cm and was aged to ~ 68 years (Neves et al. 2018). High encounter rates of this species suggest that *U. encrinus* and other sensitive cold-water corals (e.g. Nephtheid soft corals) are prone to incidental bycatch in emerging deepwater fisheries.

This study highlights the feasibility of baited cameras for use as an effective monitoring method that can be readily deployed from a variety of sampling platforms, with the ability to address critical knowledge gaps in our understanding of the biogeography of Arctic species in a region experiencing potential fisheries expansion and changing ocean conditions. BRUV surveys can provide a suite of valuable ecological data, including community composition, taxa distributions and habitat use, body size and relative and/or theoretical abundance indices. Optical surveys can generate species-level data for many organisms, or can provide insight into functional diversity where identification resolution is lacking. Depending on research objectives, these surveys may require complementary use of extractive sampling techniques (e.g. baited traps, fishing gears) to acquire voucher specimens to further identifications to species level, to quantify food-web relationships or to obtain tissue samples for molecular analyses. Future methodology should explore potential camera modifications or additions that may improve detection capacity and increase video-derived size estimates through the use of stereo-video configurations. Survey designs should be expanded to assess seasonal or annual variability in assemblage composition using cameras deployed through winter sea ice, and examine differences in effort and detectability among extractive and non-extractive sampling gears to validate the efficacy of BRUV survey methods. With adequate coverage, BRUV surveys could be a useful, non-destructive tool for exploring marine communities and detecting the presence of fishes, cold-water corals, sponges and other benthic taxa in these frontier Arctic regions.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval Deployment of baited cameras and fishing gear during this study were conducted under scientific licenses S-15/16-1041-NU, S-16/17-1017-NU and animal use protocol approvals FWI-2015-052, FWI-2016-029 granted by the Department of Fisheries and Oceans Canada and animal ethics approval by Memorial University of Newfoundland research animal care File No. 20170278.

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